



The Bucket Didn't Move the Score: A Silent Spot-Check Protocol for Auditing Self-Reported Pub-Trivia Scores, Before and After a Venue Posts an Honesty Placard

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Abstract. A pub trivia league's final score has always rested on a team's own arithmetic, checked by nobody unless a dispute is raised at the bar. We report a sixteen-week field study across twenty-two venues in a suburban trivia league, pairing each monitored team's self-reported final score against an independent, unannounced spot-check tally kept by a research assistant seated among the regular quiz-goers. Self-reported scores matched the spot-check tally, within the league's standard two-point tie-break margin, in 89.6% of 352 baseline team-nights. At the season's midpoint, fourteen of the twenty-two venues began displaying an honesty placard above the existing phone bucket; the remaining eight kept the bucket with no wording added, as a decorative-bucket control. Post-placard concordance moved to 90.9% of a further 352 team-nights, not a significant shift from baseline. Voluntary self-correction — a team amending its own submitted score downward after independently noticing an error — rose from 3.4% to 14.2% of team-nights at full-placard venues, and held flat at the decorative-bucket venues (3.1% to 3.6%). A recording-method ablation, double-tallying a subsample by paper sheet and by mechanical clicker, found the two methods statistically indistinguishable. We read the placard as changing what a team is seen to do about an error it has already caught, rather than how often one goes uncaught.

Introduction

A trivia night's final ranking depends on an arithmetic step nobody but the team performs: adding up ticks against an answer sheet, transcribing the total onto a scoresheet, and handing it to the quiz host. Where a scorekeeping error works in a team's favour, the incentive to catch it sits entirely with the team that benefits from not catching it. Most venues manage the risk with a single low-cost prop rather than a procedure: a bucket by the door for phones, on the understanding that a team without a phone in reach has one fewer way to inflate a score it already intends to report honestly. Whether the bucket, or anything posted above it, changes the accuracy of what actually gets written on the scoresheet has not, to our knowledge, been measured against an independent count of what a team's answers actually earned.

This paper reports a season-long field study built around a single design choice: an unannounced spot-check, kept by a research assistant who buys a ticket, sits with the regular crowd, and independently tallies a monitored team's correct answers without disclosing the arrangement to the team, the host, or the venue's regular staff until the season's close. Comparing that tally against the team's own submitted score separates the accuracy question from the compliance-theatre question that a posted sign more directly measures.

Three contributions follow, each stated with the caution the design supports.

- We introduce a matched spot-check design for pub trivia scorekeeping, pairing a team's self-report against an independent tally kept without the team's knowledge, across a full competitive season.

- We measure what a venue's honesty placard changes and what it does not: self-report accuracy against the spot-check, and a team's own visible act of catching and correcting its error before submission.
- We test, via a decorative-bucket control arm and a recording-method ablation, whether the placard's effect on self-correction survives once the existing phone-bucket ritual and the choice of tallying instrument are held apart from it; our results suggest, with appropriate caution, that the effect is specific to the placard's wording rather than to the bucket it sits above.

Related work

Two questions separate cleanly in the literature on unaudited self-report: whether the report is accurate, and whether the reporter is seen to behave honestly. On accuracy, self-report is a poor proxy for an independently logged behaviour across several domains once the two are compared directly: self-reported physical activity correlates only moderately with accelerometer-measured activity [1], and self-reported digital media use diverges substantially from device logs of the same behaviour [2]. Direct observation carries its own inflation: hand hygiene compliance measured under a visible auditor runs roughly three times higher than the same behaviour under unannounced electronic monitoring, a Hawthorne effect large enough to make the audit method itself part of the finding [3], consistent with Orne's earlier account of how a participant infers and obliges an experiment's implicit demand [4].

On honesty behaviour under observation, a displayed pair of eyes above an unattended honesty box nearly tripled voluntary payment relative to a neutral image in the same setting [5], and a randomised tax-audit experiment found self-reported income correcting substantially under audit threat for income an employer does not already report, with no such movement where third-party reporting made misreporting difficult regardless [6]. A large cross-country field experiment returning lost wallets found reported civic honesty tracking the money at stake in a direction opposite to a purely self-interested account [7], and identity-framed language reduced experimental cheating for gain more than behaviour-framed language aimed at the same act [8]; honor-code institutions separately

report lower self-reported cheating than institutions without one [9]. Self-officiated sport offers a closer analogue still: Ultimate Frisbee's self-refereed scoring format is reported to sustain ethical conduct without an external official [10], a pattern a later cross-divisional study of its formal scoring system corroborates [11], while aggregated non-expert judgement more broadly can converge tightly on a true value under the right conditions [12], a convergence citizen-science audits do not always reproduce [13], and one peer academic marking approximates only for holistic, not fine-grained, judgements [14].

Method

Setting and participants. The study ran across a single sixteen-week season of a suburban pub trivia league operating weekly Tuesday-night fixtures at twenty-two participating venues, each hosting an independently run quiz under a shared regional scoring format (eight rounds, ties broken by a written tiebreaker question worth up to two points). Every venue already operated a bring-your-own-team format with a phone bucket at the door, a pre-existing house rule the league did not introduce for this study.

Spot-check protocol. At each venue, on each fixture night, a research assistant purchased entry as an ordinary patron, sat within sight of two randomly pre-selected teams, and independently tallied each round's correct answers for those teams from the same publicly read answer key the venue used, without disclosing the arrangement to the teams, the host, or venue staff. The assistant's tally was sealed before the team's own scoresheet was collected, and reconciled against it only after the venue's ranking had already been announced, so the spot-check never itself corrected a team's placing. A team-night was coded **concordant** when the two totals agreed within the league's own two-point tie-break margin.

Placard rollout. At the season's midpoint (week 9), fourteen of the twenty-two venues began displaying a laminated placard above the existing phone bucket, reading "Score it straight — we spot-check," while the remaining eight retained an unmodified bucket with no added wording, as a decorative-bucket control arm (Table 1). Venues were not randomly assigned to arm, but volunteered ahead of the season by whether their operator was willing to display the

wording, a self-selection this paper returns to in Limitations.

Arm	Weeks 1–8	Weeks 9–16
Full placard (14 venues)	Bucket, no wording	Bucket + honesty placard
Decorative bucket (8 venues)	Bucket, no wording	Bucket, no wording (unchanged)

Table 1: Venue-arm allocation and rollout schedule, held with no exception sustained for more than a single fixture night.

Ground truth and self-correction. A team-night was separately coded as a **self-correction** whenever the assistant observed the team cross out and revise its own submitted total downward before handing in the scoresheet, independent of whether the revised total then matched the spot-check.

Self-Report Concordance Index. For week w 's N_w monitored team-nights,

$$SCI_w = \frac{1}{N_w} \sum_{i=1}^{N_w} \mathbb{1}[s_i - t_i \leq 2],$$

where s_i is the team's self-reported total and t_i the assistant's independent tally for team-night i ; Figure 1 reports SCI_w across all sixteen weeks.

Recording-method ablation. For a randomly drawn subsample of 120 team-nights spread evenly across the season, the assistant recorded the same round-by-round tally simultaneously by paper tally sheet and by a silent mechanical clicker, to test whether the recording instrument itself shifted the resulting discrepancy.

Results

Concordance holds flat across the placard rollout. Figure 1 plots the weekly Self-Report Concordance Index across all 704 monitored team-nights. Baseline concordance (weeks 1–8, $N = 352$) averaged 89.6%; post-placard concordance (weeks 9–16, $N = 352$, pooled across both arms) averaged 90.9%, a difference that does not reach significance ($\chi^2(1) = 0.31$, $p = .58$). The index shows no visible step at week 9, consistent with self-report accuracy having already been high before any placard was displayed.

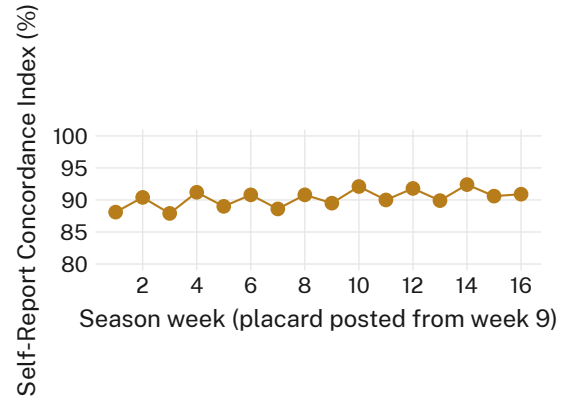


Figure 1: Weekly self-report concordance held at 89.6% before the week-9 placard rollout and 90.9% after it (n.s.).

Self-correction rises only where the placard was posted. Figure 2 compares the self-correction rate before and after week 9, split by arm. At the fourteen full-placard venues, self-correction rose from 3.4% to 14.2% of team-nights ($\chi^2(1) = 18.6$, $p < .001$); at the eight decorative-bucket venues, it moved from 3.1% to 3.6% over the same span (n.s.). The pre-existing phone bucket, present at every venue throughout the season, does not itself appear to have driven the rise; the placard's wording, layered onto an otherwise unchanged bucket, does.

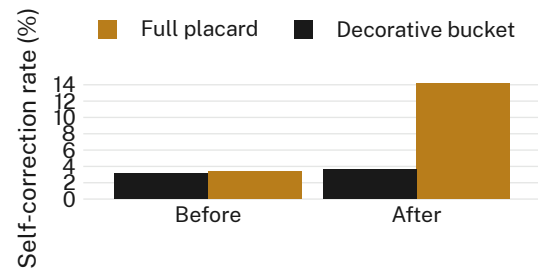


Figure 2: Self-correction rate rose from 3.4% to 14.2% of team-nights at full-placard venues and held flat at decorative-bucket venues.

Recording method changes nothing. Figure 3 compares the distribution of absolute point discrepancy between self-report and spot-check across the 120 double-recorded team-nights, split by whether the assistant's tally for that comparison was drawn from the paper sheet or the clicker. The two distributions are close to indistinguishable (medians of 1 point either way; Mann-Whitney $U = 1742$, $p = .81$); we retain the paper tally sheet as the season's primary instrument for its lower failure rate under poor pub

lighting, though the ablation gives no measured reason to prefer it on accuracy grounds.

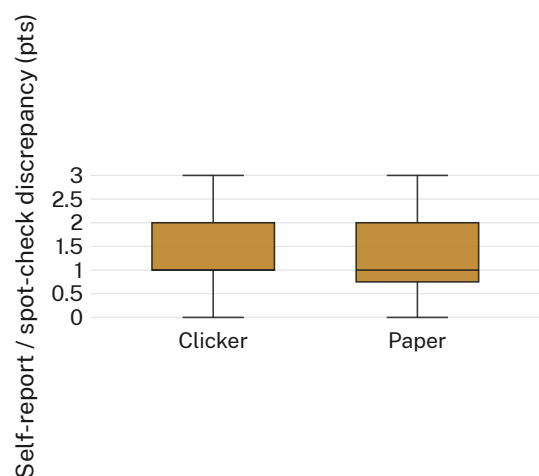


Figure 3: Self-report / spot-check discrepancy is statistically indistinguishable between paper-sheet and clicker recording (n.s.).

Discussion

Read together, the three results separate an accuracy question from a compliance-theatre question that a posted placard more naturally answers. Self-report was already close to the spot-check tally before any placard existed, leaving comparatively little room for a sign to move a concordance index that a season of data suggests was never badly wrong to begin with. What moved was a team's visible response to an error it had already caught on its own: more likely, once told a spot-check was in play, to be crossed out and corrected on the scoresheet rather than submitted and left. A comparable rollout elsewhere at the University found a compliance measure climb sharply once a disposal system's label went live, while the dispute rate underneath it held flat across the same window, a pattern this season's data recalls closely [15]. A parallel finding on a posted safety cue found reported trust in the cue sliding only slightly as it grew stale, while the checking behaviour behind it moved substantially further over the same span [16]; our placard, on this reading, moved what a team is seen to do about its own arithmetic rather than the arithmetic itself. Our results suggest, with the appropriate caution, that a league wanting fewer disputed scores at season's end already has most of the accuracy it needs before it prints anything.

Limitations

Venues were not randomly assigned to the full-placard or decorative-bucket arm; an operator's willingness to display the wording may correlate with qualities of the venue's existing trivia culture that this design cannot separate from the placard's own effect, though the consistency of the concordance null across both arms limits how much of that confound could plausibly be driving it. The spot-check covered two teams per venue per night rather than the full field, a sampling choice that kept the assistant's cover plausible at the cost of season-wide coverage; the teams sampled were rotated weekly and not self-selected, which we consider the more consequential safeguard. Self-correction was coded from a single assistant's visible observation of the scoresheet, a measure we could not blind further without losing the behaviour it was recording.

Conclusion

Across a season of spot-checked pub trivia scoring at twenty-two venues, an honesty placard posted above an existing phone bucket left the accuracy of what teams reported about themselves essentially where it already was, and changed instead how visibly a team dealt with an error once it had already found one. The distinction matters for any venue treating a rising self-correction count as evidence that its scoring has become more honest, when the more defensible reading is that scoring was already honest enough for the placard to have little arithmetic left to improve. Future work should test whether the same placard, posted for a full season rather than from its midpoint, changes the rate at which an error is caught in the first place, not only what a team does once it has been.

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